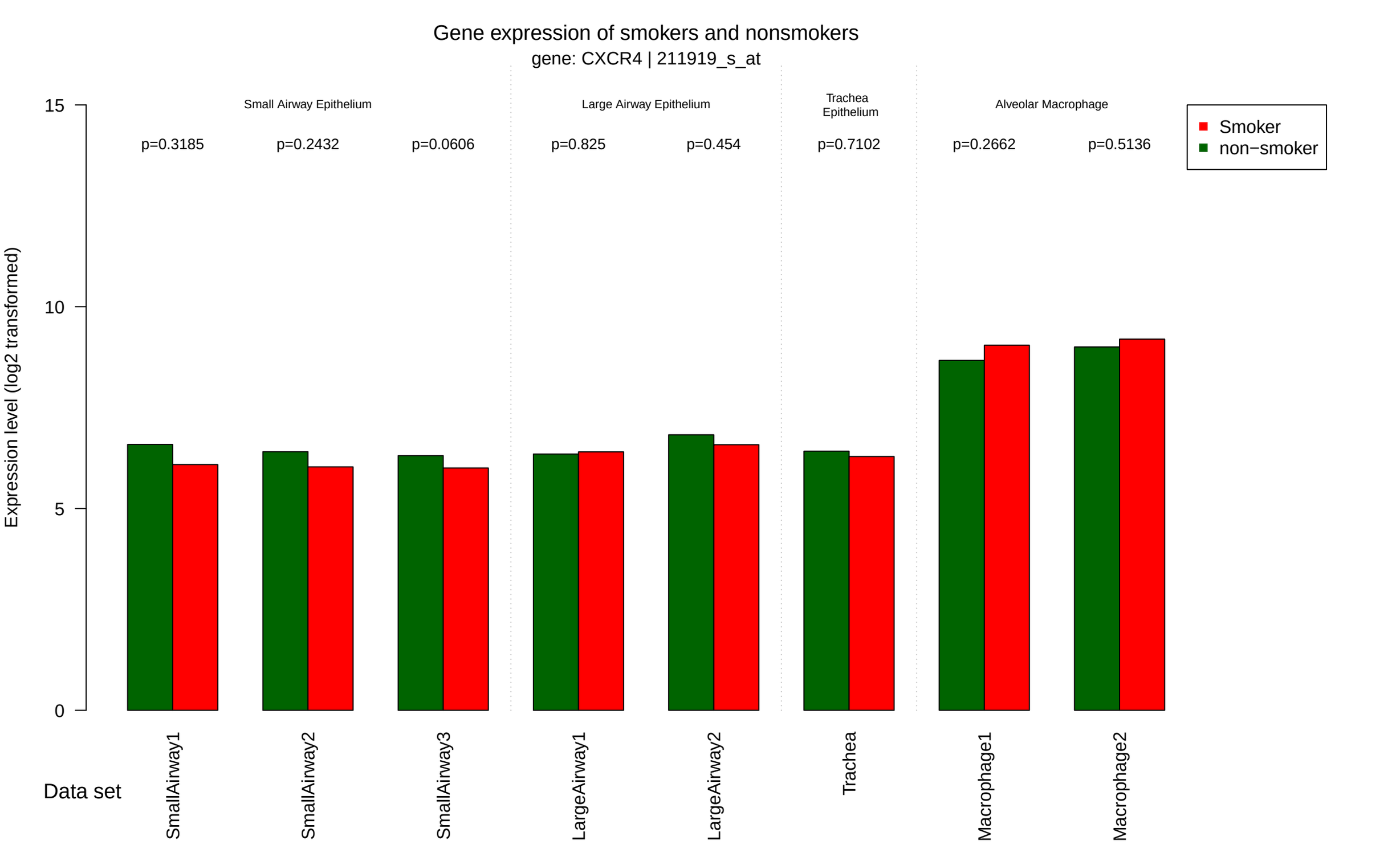
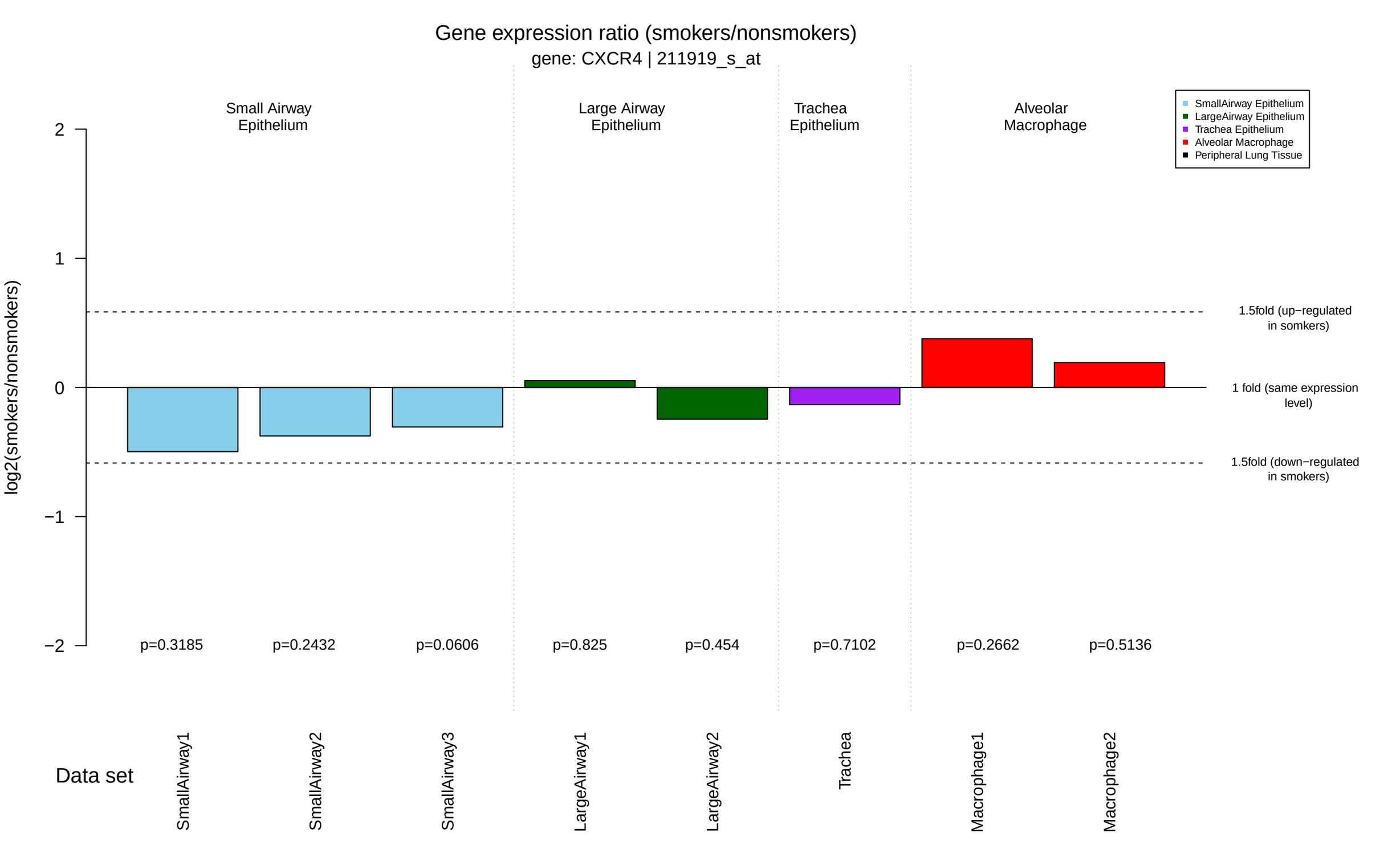
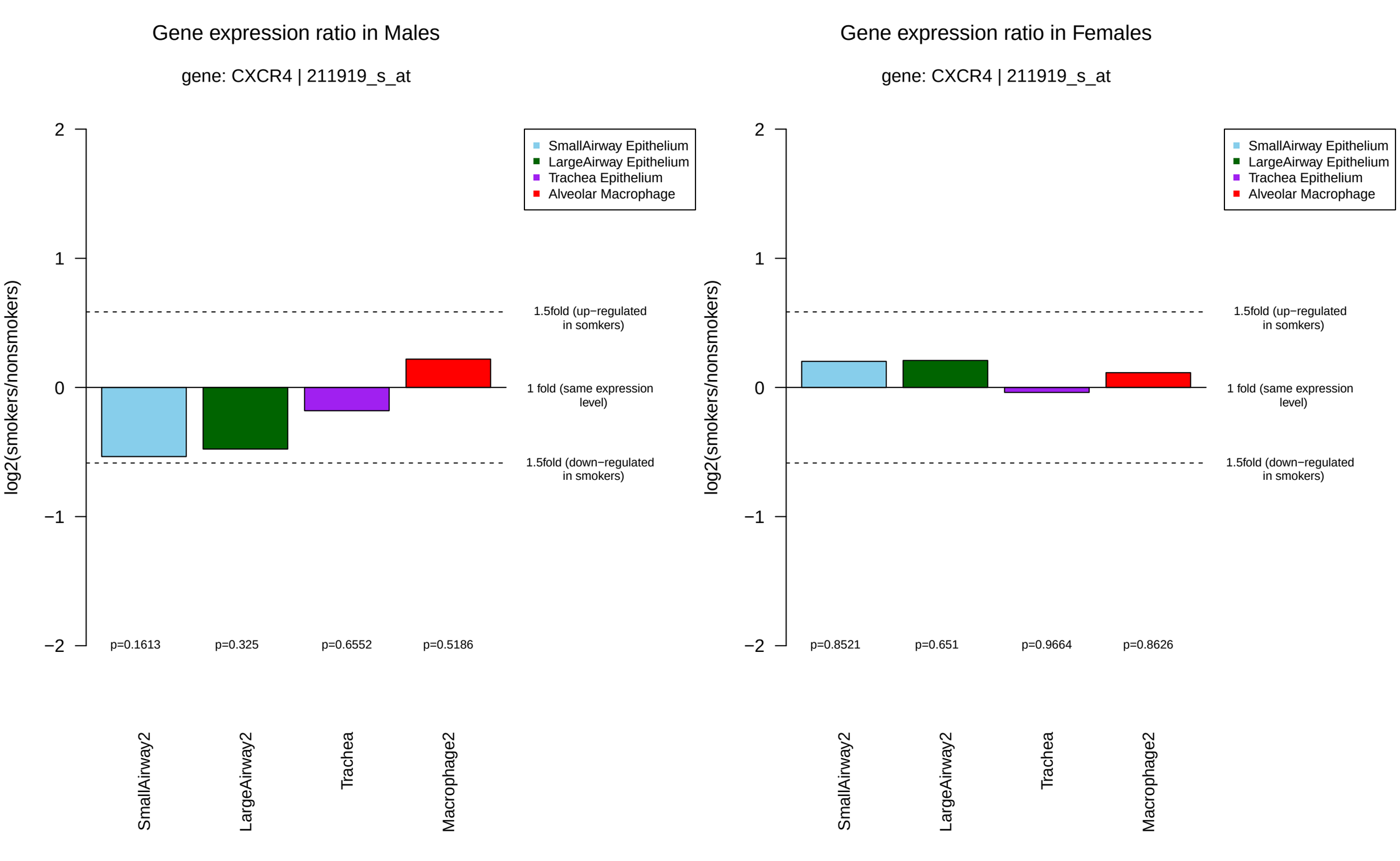
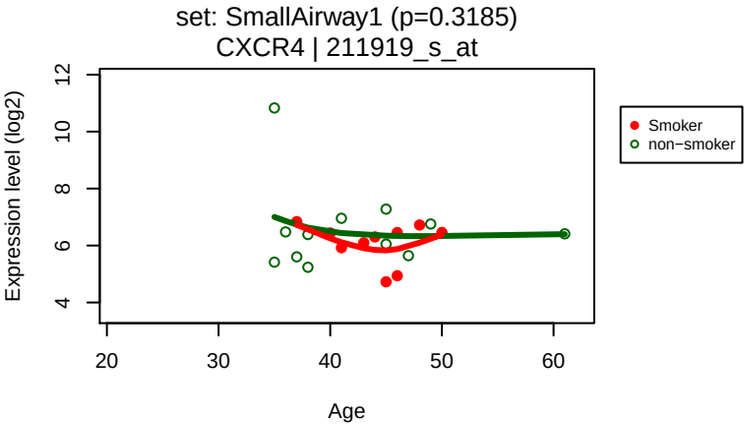
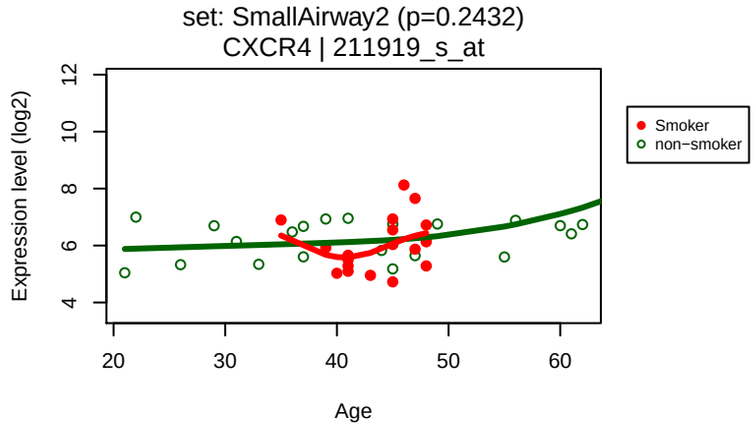
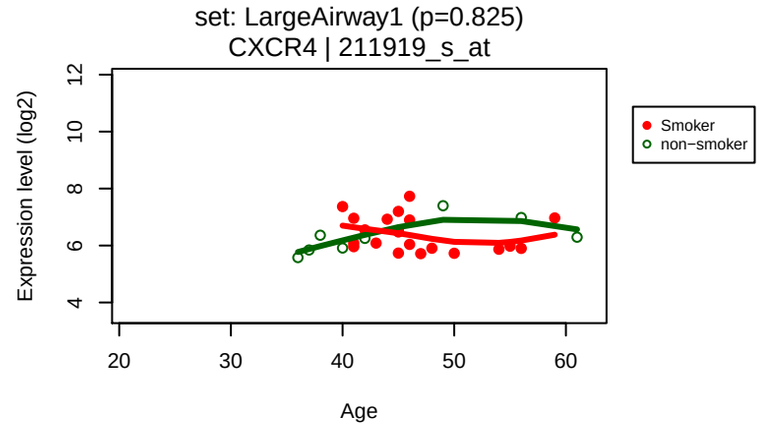

Gene expression of smokers and nonsmokers
gene: CXCR4 | 211919_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3185
p=0.2432
p=0.0606
p=0.825
p=0.454
p=0.7102
p=0.2662
p=0.5136
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

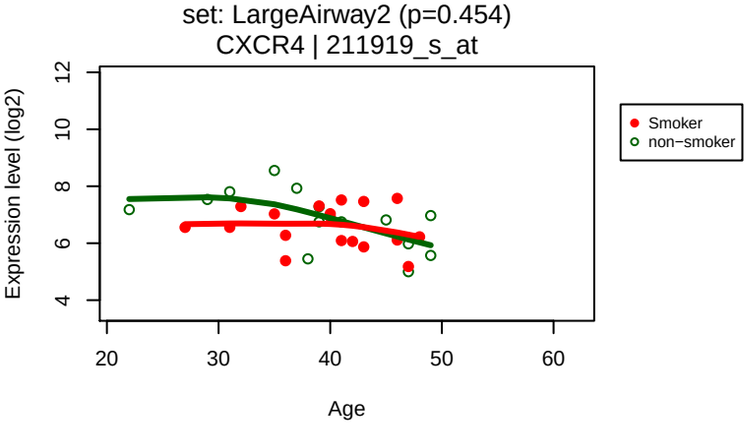

Gene expression ratio (smokers/nonsmokers)
gene: CXCR4 | 211919_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3185
p=0.2432
p=0.0606
p=0.825
p=0.454
p=0.7102
p=0.2662
p=0.5136
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

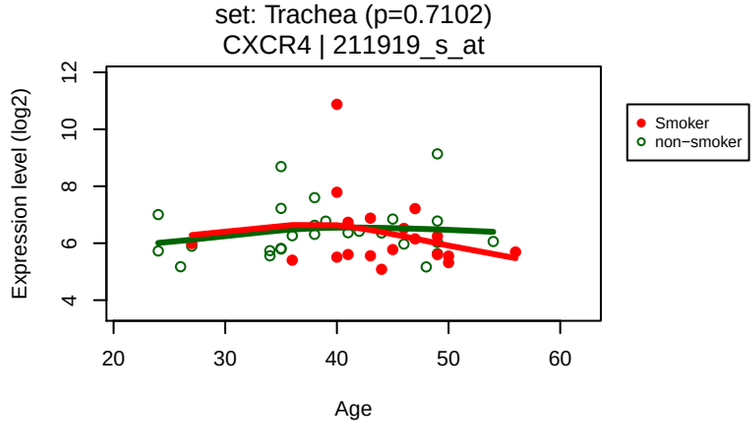

Gene expression ratio in Males
gene: CXCR4 | 211919_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1613
p=0.325
p=0.6552
p=0.5186
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CXCR4 | 211919_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8521
p=0.651
p=0.9664
p=0.8626
SmallAirway2
LargeAirway2
Trachea
Macrophage2

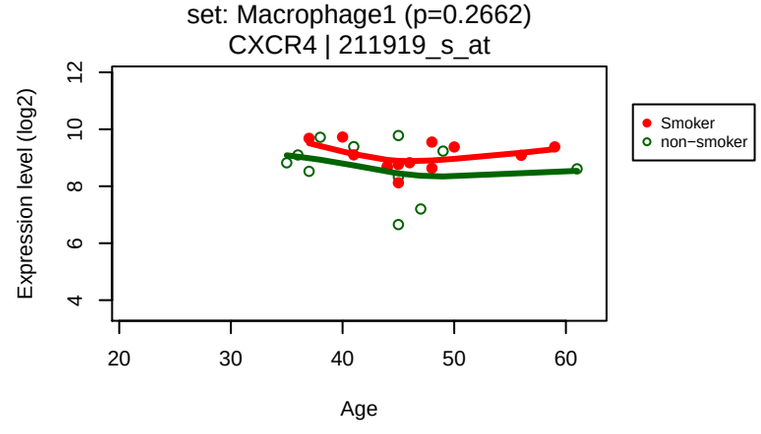

set: SmallAirway1 (p=0.3185)
CXCR4 | 211919_s_at
Expression level (log2)
Age
Smoker
non−smoker

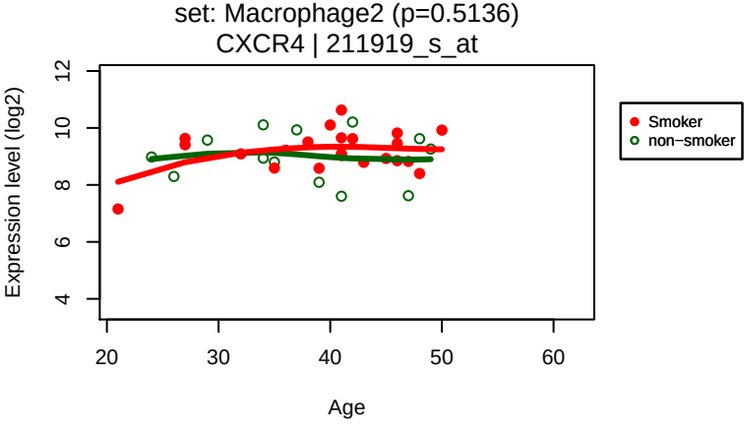

set: SmallAirway2 (p=0.2432)
CXCR4 | 211919_s_at
Expression level (log2)
Age
Smoker
non−smoker

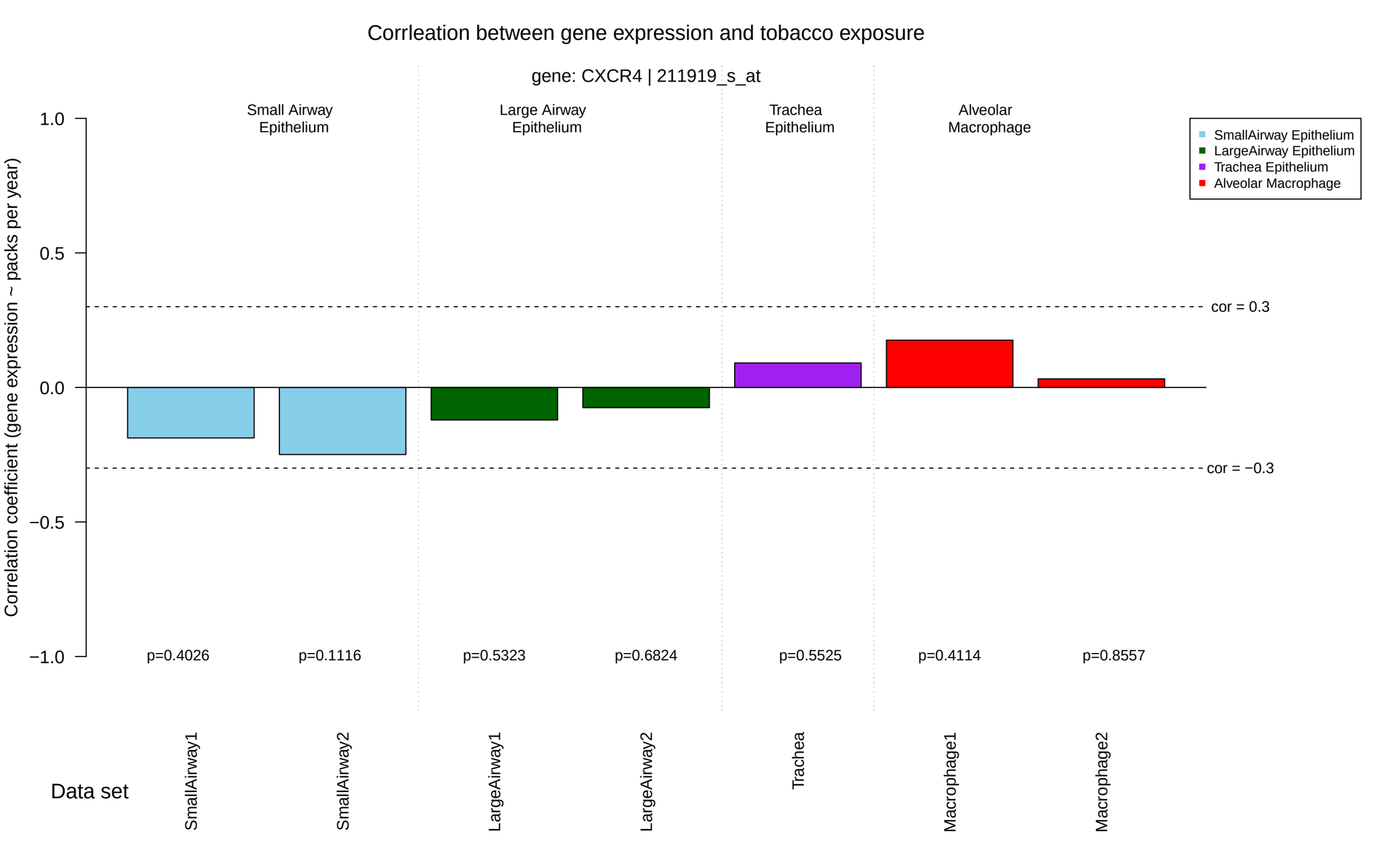

set: LargeAirway1 (p=0.825)
CXCR4 | 211919_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.454)
CXCR4 | 211919_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7102)
CXCR4 | 211919_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2662)
CXCR4 | 211919_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5136)
CXCR4 | 211919_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CXCR4 | 211919_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4026
p=0.1116
p=0.5323
p=0.6824
p=0.5525
p=0.4114
p=0.8557
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2